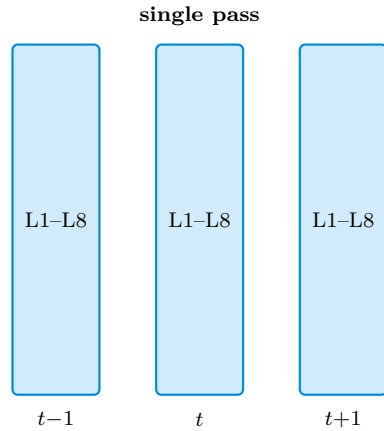


Training-time information flows (functional blocks)

Depth recurrence OFF

Vanilla transformer

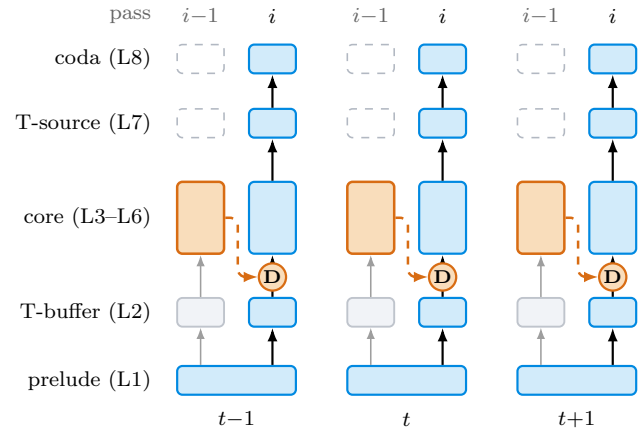
One pass through L1–L8 at every position (embedding and head not shown).



Depth recurrence ON

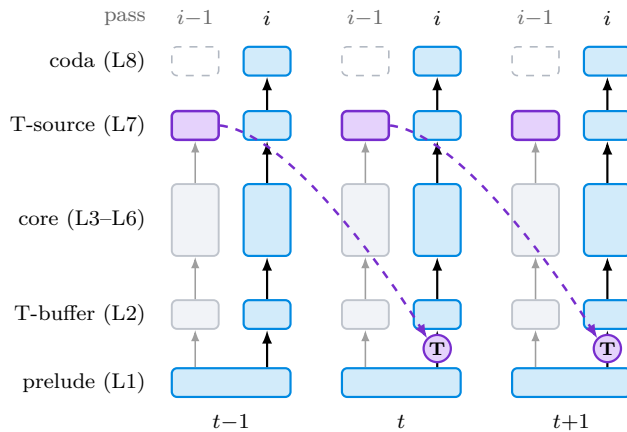
Depth-recurrent / looped LM

Pass $i-1$ writes the core output as depth state; pass i reads it at the same position, just before the core.



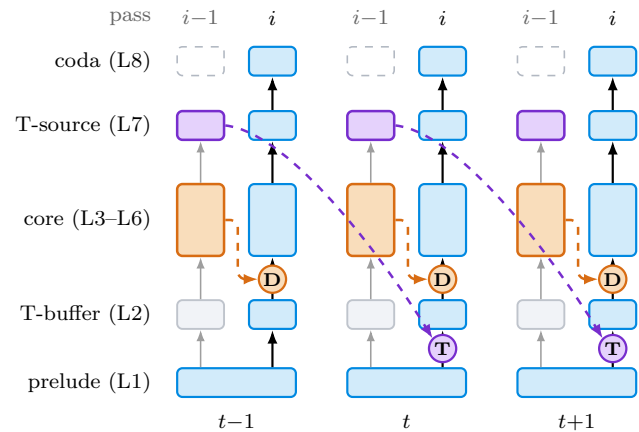
Temporally recurrent

Pass $i-1$ writes the T-source output (L7) as temporal state; pass i reads it one position later, just after the prelude.



Depth + temporal hybrid

Both at once: depth state stays at its position (short orange arrows), temporal state moves one position forward (long purple arrows).



pass $i-1$ block

pass i block

not run on this pass

prelude: runs once

writes depth state

writes temporal state

depth read (same position)

temporal read (next position)